equivalent. Thus it is that, when studying gravity in the solar system, physicists often switch back and forth with impunity between the Newtonian paradigm, the curved spacetime paradigm, and also the flat spacetime paradigm, using at any time whichever one strikes their fancy or seems the more insightful.[1]

Sometimes people new to a field of research are more open-minded than the old hands. Such was the case in the 1970s, when new people had insights that led to a new paradigm for black holes, the *membrane paradigm.*

In 1971 Richard Hanni, an undergraduate at Princeton University, together with Remo Ruffini, a postdoc, noticed that a black hole's horizon can behave somewhat like an electrically conducting sphere. To understand this peculiar behavior, recall that a positively charged metal pellet carries an electric field which repels protons but attracts electrons. The pellet's electric field can be described by field lines, analogous to those of a magnetic field. The electric field lines point in the direction of the force that the field exerts on a proton (and oppositely to the force exerted on an electron), and the density of field lines is proportional to the strength of the force. If the pellet is alone in flat spacetime, its electric field lines point radially outward (Figure 11.3a). Correspondingly, the electric force on a proton points radially away from the pellet, and since the density of field lines decreases inversely with the square of the distance from the pellet, the electric force on a proton also decreases inversely with the square of the distance.

Now bring the pellet close to a metal sphere (Figure 11.3b). The sphere's metal surface is made of electrons that can move about on the sphere freely, and positively charged ions that cannot. The pellet's electric field pulls a number of the sphere's electrons into the pellet's vicinity, leaving excess ions everywhere else on the sphere; in other words, it *polarizes*[2] the sphere.